

Reinforcement Learning Approach to Adaptive Traffic Signal Control using SUMO-RL

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Abstract—Traffic control in urban environments is a complex and critical challenge, with growing urbanisation and increasing vehicular congestion necessitating innovative solutions. Reinforcement learning has emerged as a promising approach to optimise traffic control and improve overall transportation efficiency, thanks to its ability to grasp the intricacies of complex issues without human involvement autonomously. In this paper, we propose a modified version of the SUMO-RL package and demonstrate its advantages in the context of our ongoing research regarding traffic control in the city of Žilina. The proposed modifications include support for custom reward functions for different agents and custom observation functions, representing a crucial design component when solving traffic signal control as a multi-agent reinforcement learning problem. We also present our preliminary results, which demonstrate the potential of reinforcement learning in addressing traffic control challenges.

Index Terms—reinforcement learning, SUMO-RL, control, simulation, traffic, modelling

I. INTRODUCTION

In modern urban environments, intersections often serve as the focal points of traffic congestion, causing vehicles to come to a halt and resulting in significant delays for commuters [1], [2]. This not only leads to frustration and wasted time but also negatively impacts the overall quality of life in cities.

Apart from these adverse effects, intersections contribute significantly to air pollution and increased fuel consumption

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[3]. Prolonged idling at intersections releases harmful pollutants such as carbon monoxide and nitrogen oxides, which have adverse health effects and contribute to environmental degradation [4]. Inefficient intersection control, leading to traffic congestion, also imposes a substantial economic burden. The excessive time spent in traffic not only increases fuel consumption but also reduces productivity, contributing to economic losses [3].

Intelligent traffic light control is emerging as a viable method for mitigating traffic congestion. To address many challenges present in intelligent traffic light control, we propose modifications to the existing SUMO-RL [5] package to enhance its capabilities in RL domain. The initial essential adjustment pertains to enabling support for the agent's reward functions. Another crucial package modification involves facilitating the incorporation of custom observations.

This paper is organised as follows. In section II, we discuss the related work. Proposed modifications to the SUMO-RL package are proposed in section III. In section IV, we showcase experimental outcomes. We discuss our methods in section V. Finally, we conclude this paper in section VI.

II. RELATED WORK

In recent years, the field of transportation engineering and traffic control has witnessed a growing interest in the integration of artificial intelligence and reinforcement learning (RL) techniques to enhance the efficiency of traffic control systems [6]–[11]. The optimisation of traffic signal control has become a prominent research area due to the increasing challenges associated with urban congestion and the demand

for sustainable, eco-friendly transportation solutions [12], [13]. Several prior works have explored the application of reinforcement learning algorithms in the context of smart traffic control systems [14]–[16]. These studies have sought to address issues such as reducing traffic congestion, minimising travel times, decreasing emissions, and improving overall transportation network performance [17].

Given the diverse and intricate nature of urban traffic conditions, the application of reinforcement learning to alleviate traffic congestion has emerged as a prominent and promising area of research. In particular, effectively coordinating traffic lights at a number of intersections stands out as a significant challenge in the realm of multi-agent reinforcement learning (MARL) [18]. The majority of prior studies in MARL relied predominantly on traditional Q-learning approaches. In Q-learning, an iterative process oscillates between exploration, during which the agent experiments with novel actions to uncover superior strategies, and exploitation, where the agent leverages its existing knowledge to choose actions aimed at maximising immediate rewards [19], [20].

Many small and large cities, even in developed countries, continue to employ outdated fixed-time signal control strategies, often with suboptimal optimization and maintenance [21], [22]. These control strategies utilize fixed time intervals for all green phases, determined according to historical traffic demand, without accounting for potential fluctuations in traffic demand. Hence, it fails to achieve the global optimum. The rising interest in enhancing transportation systems through intelligent means has led to a surge in research focusing on the application of machine learning techniques, particularly reinforcement learning, to optimize traffic signal control. These techniques are appealing because they can autonomously uncover effective control strategies for intricate tasks, like traffic control. These tasks pose challenges in computing optimal solutions directly and are difficult to address with manually programmed solutions [14]. Numerous recent academic publications delve into this field of study [23]–[25]. RL learns through trial and error, needing many interactions before convergence, incurring real costs like traffic congestion and accidents. Hence, most RL methods primarily experiment in simulators because they generate data more affordably and swiftly than real-world trials. Differences between simulation and reality limit the practical application of RL in real-world settings, despite its significant success in domains such as games [21]. Previous papers have explored the literature on traffic control and reinforcement learning and also describing the fundamentals of traffic signal control, RL algorithms and approaches [25]–[27].

Additionally, in urban settings, coordinating the timing of adjacent traffic signals is essential as they are often situated close together. This process, known as signal timing coordination, must be done collectively to prevent decisions made at one signal from negatively impacting traffic flow at another [24]. Understanding this broader context is essential for appreciating the challenges and opportunities inherent in our research endeavor. By leveraging insights from existing

literature, we aim to build upon prior work and contribute novel solutions that address the complexities of real-world traffic control problems in the realm of RL.

III. MATERIALS AND METHODS

SUMO-RL offers the necessary tools for researching traffic light signal (TLS) timing control. SUMO-RL interacts with SUMO through the TraCI interface [28] and brings a simple interface to work with reinforcement learning for traffic signal control, supports multiagent RL, and several interfaces including the `gymnasium.Env` [29] (a de-facto standard for single-agent RL) and PettingZoo [30] (a similar standard for multi-agent RL) [5].

One of the notable shortcomings of SUMO-RL in the past was its reliance on a single unified representation with a fixed observation function and reward function. This limitation was considerable because the observation function and the reward function collectively make up a substantial portion of the design space in multi-agent traffic control – together with deep architectures and learning methods, this is where most of the innovation happens.

For this reason, we propose several related extensions to SUMO-RL, which have now been incorporated into upstream and which are detailed in the following subsections. The changes now enable us to establish an environment with our own observations and our own observation function, thus facilitating a deeper understanding and analysis of the subject matter at hand. Next, we present a case study on the city of Žilina to demonstrate how RL can be utilised in practice.

A. Support for Different Reward functions for Different Agents

Our first proposed modification lies in the reward function domain. The original state of the SUMO-RL package only made it possible to define a single common reward function for all agents – and even that with limited flexibility. With our changes, when creating an environment object, it is possible to independently specify the reward function for each agent (in the form of a dictionary, keyed by agents’ identifiers; the identifiers are generally tied to particular intersections). It is important to emphasize that for a custom reward function to grant rewards based on the agent’s observations, it must include at least one argument: the `TrafficSignal` object. This object provides the reward function with crucial data regarding the current state of the environment. Additionally, a registration function and a validation check for the specified reward function (error handling) have also been added [31], [32].

B. Support for Custom Observation Functions

Another essential package modification involves the introduction of support for custom observations. The package was originally designed to function with only a single, hard-coded observation format. In the updated version, when creating a `SumoEnvironment` object, users now have the flexibility to define their custom observation format by deriving from the `ObservationFunction` class [33] –

this allows researchers to easily experiment with different observation formats – we already rely on this functionality in a follow-up project, where the goal is to train a more generalist agent, instances of which can be applied across a number of different traffic scenarios, intersection topologies, etc. – and the ability to manipulate the observation format is crucial. Finally, SUMO-RL’s original observation function is, of course, still included as the default option (class `DefaultObservationFunction`).

C. Utilisation of Custom Reward Functions and Observation Functions

In summary, the SUMO-RL package had a few limitations that we have addressed one by one, which enabled us to design an environment with custom observations, inspired by a representation found in the RESCO package [8] – namely the `drq_norm` and `wait_norm` methods used by the IDQN agent. Bringing this (and potentially other similar) setups into SUMO-RL allows us to setup more dynamic simulation scenarios. RESCO works with static pre-captured datasets, which are valuable in the sense that they enable benchmarking on real data, but not so useful for training, where a lot of varied data is needed. By incorporating custom observation and reward functions into SUMO-RL, we provide researchers with flexibility to experiment with different setups and to generate varied traffic data on the fly without the need to rely on static datasets [34].

D. Map of Žilina

Conducting traffic simulations in a local city offers significant advantages, providing a context-specific and highly relevant environment for research and analysis. Firstly, familiarity with the local infrastructure, road layouts, and traffic patterns allows for a more accurate representation of real-world conditions. This authenticity is crucial for developing effective traffic control solutions that resonate with the specific challenges and dynamics of our community. We have created a map for the city centre of Žilina, as shown in Fig. 1, while a satellite image of this area can be seen in Fig. 2.

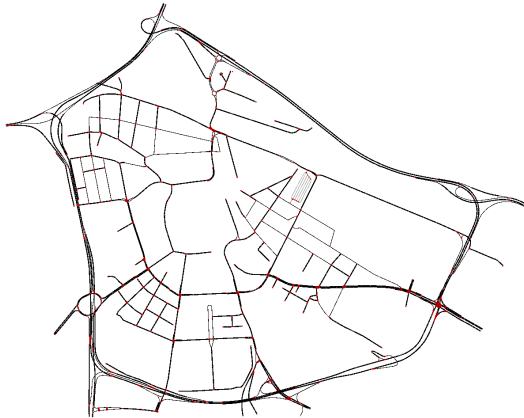


Fig. 1. Map of the city centre of Žilina.

The foundation of this map is the cartographic data obtained from OpenStreetMap, known for being a versatile source of geographic information. It is crucial to highlight that the raw data from OpenStreetMap underwent multiple updates, aiming to ensure the utmost accuracy and relevance for our study. During this detailed data refinement process, redundant information was removed, attributes were standardised, and the precision of intersection and traffic signal layouts was meticulously verified. We implemented a range of modifications, including reshaping intersections, driving lanes, and turning lanes. Additionally, we altered the flow of one-way streets, adjusted routes for public transport, imposed restrictions on vehicle entry into specific areas, and eliminated residential zones where vehicles were primarily parked.



Fig. 2. Satellite image of the city centre of Žilina [35].

E. Training

Training a reinforcement learning model is a challenging task that requires finding the right balance between exploration (trying out new actions to discover better strategies) and exploitation (leveraging learned knowledge to make optimal decisions).

The training of our reinforcement learning model, utilising an AMD Ryzen 7 5700X CPU at 4.60GHz, NVIDIA GeForce RTX 3080 10GB, and 2x Kingston DDR4-3200 CL16 16GB of RAM, yielded promising results.

To select appropriate learning parameters, it is generally beneficial to conduct multiple runs of the program with different parameter combinations and assess the learning progress.

IV. RESULTS

The most reliable measure for assessing whether a system is acquiring knowledge lies in monitoring the progression of received rewards over the learning phase. Ideally, this reward curve should exhibit a clear upward trend, even though there is generally also a fair amount of noise related to the stochasticity associated both with the environment dynamic and with the agent’s exploration policy. Exploration in particular can be expected to temporarily result in lower immediate rewards.

Our learning outcomes are shown in Fig. 3, 4, 5, and 6, as well as summarized in TABLE I. This table compares various traffic signal control algorithms, including the IDQN approach using the Perceiver architecture, IPPO, Actuated TLS, and Fixed Time, using average trip time, average delay, average wait, and average queue length as performance indicators in seconds/number of vehicles.

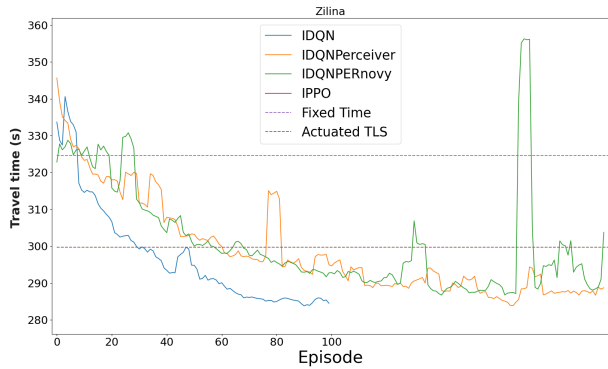


Fig. 3. This graph shows the average travel time in seconds for each episode for different types of TLS control.

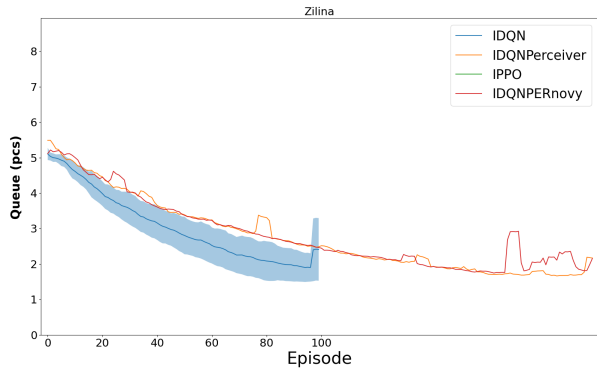


Fig. 4. This graph shows the average queue length for each episode for different types of TLS control.

TABLE I
AVERAGE RESULTS FOR DIFFERENT ALGORITHMS.

	Delay	Trip Time	Wait	Queue
IDQN	72.68	281.44	24.46	1.86
IDQNPerceiver	73.81	283.11	25.2	1.61
IPPO	95.72	285.15	42.07	3.49
Fixed Time	106.03	303.07	53.21	-
Actuated TLS	81.94	324.62	32.3	-
IDQNPERnovy	76.51	299.67	26.83	1.71

For these experiments, 100 episodes were used. For performance measuring, we use the average trip time of all vehicles on the map, the average wait times across all vehicles on the map, the average delay and the average queue length.

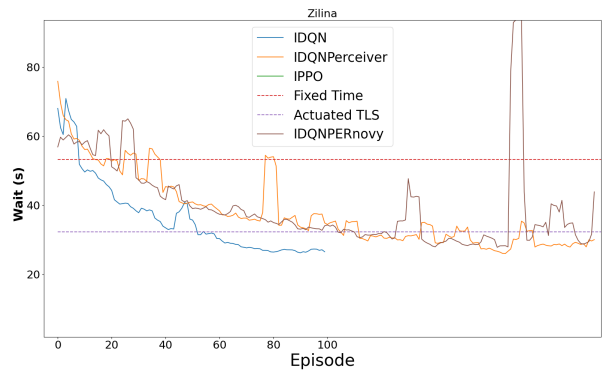


Fig. 5. This graph shows the average wait time in seconds for each episode for different types of TLS control.

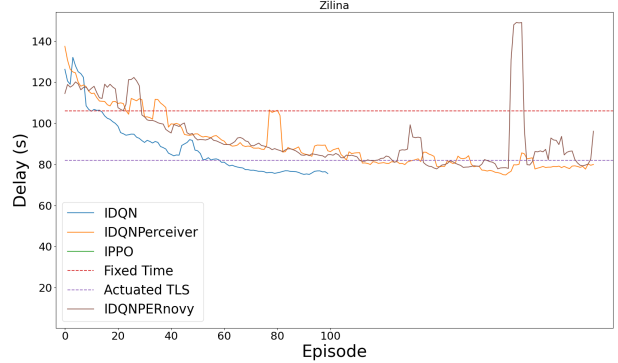


Fig. 6. This graph shows the average trip time in seconds for each episode for different types of TLS control.

Average trip time is the amount of time between when cars first appear on maps and when they vanish. Average wait time means cars waiting on the maps while their speed is zero. The average queue represents the number of vehicles waiting in the queue. Average wait time presents the amount of time the car unwillingly spent waiting.

In terms of average trip time, IDQN demonstrates the lowest value of 281.44 seconds, indicating that this algorithm is successful in reducing overall travel times for vehicles passing through the controlled intersections. The IDQNPerceiver and IPPO exhibit a slight lag, with IDQNPERnovy averaging 299.67 seconds, indicating a performance deficit. Despite its intentional design with increased depth compared to other networks, this architecture fails to deliver satisfactory results. Notably, when compared to conventional TLS methods such as Fixed Time and Actuated TLS, all alternative approaches outperform it.

It is important to note that the IDQN curve does not extend to the end due to saturation. Even with a larger number of episodes, it did not yield significantly better results. The other methods required at least 250 episodes to reach saturation, and the results did not show improvement. The hyperparameters

used by the perceiver can be found in the Table II.

The Fixed Time Controller and Actuated TLS were extracted from SUMO using the tripinfo command. For Fixed Time, a cycle of 90 seconds was set, while the Activated TLS had a range between 5 to 60 seconds.

TABLE II
PERCEIVER HYPERPARAMETERS

Parameters	Value
input_channels	1
input_axis	2
num_freq_bands	5
max_freq	7.49
depth	6
num_latents	34
latent_dim	32
cross_heads	1
latent_heads	8
cross_dim_head	64
latent_dim_head	64
num_classes	act_space
attn_dropout	0.293
ff_dropout	0.447
weight_tie_layers	False
fourier_encode_data	True
self_per_cross_attn	2

V. DISCUSSION

The research presented in this paper focuses on simulations conducted within the specific urban context of Zilina. While the findings offer valuable insights into the dynamics of urban systems within this particular city, it is imperative to address the question of generalizability to other urban environments. Urban landscapes vary significantly in terms of infrastructure, population density, socio-economic factors, and cultural characteristics, among other variables.

Achieving improved outcomes in the domain of deep learning is a multifaceted challenge, often lacking a definitive, one-size-fits-all solution. The complexity of this task is further compounded by the presence of multi-agent environments.

An alternative approach involves exploring an optimal neural network architecture that encapsulates the strategy's parameters. As previously discussed, while the input and output layers are determined by the environment, the hidden layers remain customisable. Various configurations of hidden layers were experimentally tested. However, determining the most effective configuration is challenging, primarily due to the limited number of learning steps taken.

The primary objective of this paper was to highlight the potential for resolving traffic control issues by employing deep learning techniques incorporating rewards within simulation scenarios. In the course of our research, we encountered various challenges, including the adaptation of the existing interface to simulate a multi-agent environment, the integration of this environment into a deep learning application with a reward system, and the identification of appropriate input parameters for the learning process.

The output of this paper consists primarily of a revised SUMO-RL interface (where our enhancements have been officially incorporated into the package, making them accessible to other researchers), a well-prepared deep learning environment for SUMO scenarios with reward mechanisms, and potential strategies to enhance learning results within this environment. The primary modification involved enhancing support for the agents' reward functions. This feature enables the specification of distinct reward functions for each of TrafficSignal objects independently. This enhances flexibility and enables finer control over the reinforcement learning process for individual agents. Another proposed modification is to support custom observations. As noted, the original package included only one predefined observation type, which was embedded within the TrafficSignal object itself. This modification facilitates researchers in effortlessly experimenting with various forms of observations.

Our ongoing research is aimed towards our city of Žilina. It might facilitate a collaboration with local authorities and stakeholders, fostering a more seamless integration of research findings into practical applications. By conducting traffic simulations in the environment, the research becomes not only academically robust but also practically impactful, directly addressing the unique demands of our local transportation systems.

The issue of generalising the findings of this paper requires further study through the comparison of simulations in various cities. This will help us understand how relevant and effective the patterns and strategies identified in the city of Zilina are in different urban areas. Hence, future research should aim to incorporate simulations from diverse urban environments to enhance the discussion on the applicability of the findings beyond the city of Zilina.

Additionally, training reinforcement learning algorithms can be computationally expensive due to the iterative trial-and-error learning process, complex neural network architectures, and the exploration-exploitation trade-off. This necessitates running numerous simulations or episodes over extended periods, requiring significant computational resources and time. Overall, the computational expense and resource requirements of training reinforcement learning algorithms pose significant challenges to researchers and practitioners alike.

To sum up, intelligently managing traffic lights with reinforcement learning is crucial for combating congestion in urban areas. By dynamically adapting to changing conditions, such systems can optimise traffic flow, reduce delays, enhance safety, and minimise environmental impact. With scalability and continuous improvement, they might offer a flexible and effective solution for improving urban transportation networks in the near future.

VI. CONCLUSIONS

In conclusion, the integration of reinforcement learning techniques within the SUMO-RL has shown significant promise and potential in addressing complex transportation and traffic control challenges. SUMO-RL offers a powerful

tool for improving urban mobility, and its continued development and refinement hold the potential to revolutionise the way we manage traffic and transportation systems in the future. This paper underscores the importance of ongoing research and development in this field and the necessity of addressing the unique challenges presented by complex urban environments. Furthermore, it presents valuable enhancements to the existing SUMO-RL package. Lastly, our ongoing research in the city center of Žilina is presented.

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